UTM UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2024/2025

Subject	:	PROBABILITY & STATISTICAL DATA ANALYSIS (SCST 1223)
Task	:	Chapter 1 - Chapter 4
Due	:	**Week 7 (30 April 2025)** *During class lecture only!*

INSTRUCTION:

1. This is an assignment for a **GROUP of 3-4**. Please clearly write the group members' name & matric number on the front page of the submission.
2. This assignment will give 5% of the overall course marks.
3. Only **HANDWRITTEN** submission is accepted:
    a. Submissions using any reporting or statistical tools (e.g.: MS Word, MS Excel, etc,) will be **REJECTED**.
    b. Make sure the submission is neatly written. Any submission with handwriting that is unreadable will be **REJECTED**.
    c. For answer that need to draw graphs, using graph paper is optional. You can use plain paper.
    d. Round your answers to **TWO** decimal places.
    e. Please scan/snapshot your work and save as a PDF file or submit a hard copy to the lecturer.
4. Submission:
    a. Only **ONE** group member needs to submit on behalf of the group.
    b. HARDCOPY FORM ONLY.

QUESTION 1 **[15 MARKS]**

For each of the following statements, identify the correct level of measurement: Nominal, Ordinal, Interval, or Ratio.

No.	Statement	Level of Measurement
1	Movie genres like Action, Comedy, Horror.	
2	Customer satisfaction: Unsatisfied, Neutral, Satisfied.	
3	Years on a calendar (e.g., 1990, 2000, 2020).	
4	Distance (in meters) ran during a school sports day.	
5	Age of participants in a health survey.	
6	Nationalities of tourists.	
7	Hotel star ratings (1 to 5 stars).	
8	Time (in minutes) taken to complete a puzzle.	
9	Types of animals at a zoo.	
10	Ranking of top universities.	
11	Number of books read in a month.	
12	Education level: High School, Diploma, Degree.	
13	Car brands: Toyota, Honda, Proton.	
14	Level of agreement: Strongly disagree to Strongly agree.	
15	Speed (in km/h) of different vehicles.	

QUESTION 2 **[20 MARKS]**

a) The dataset below shows the weekly expenses (in RM) of 11 Faculty of Computing students:

85, 90, 120, 100, 90, 110, 95, 90, 105, 115, 100

Calculate the mean, median, and mode of the dataset above.

(10 Marks)

b) Table 1 shows the waiting time of customers in a bank.

Table 1: Waiting Time (minutes)

Class Interval	Frequency
2.5 – 5.5	6
5.5 – 8.5	10
8.5 – 11.5	22
11.5 – 14.5	15
14.5 – 17.5	5
17.5 – 20.5	2

Based on Table 1, find the:

i) Mean

(4 Marks)

ii) Median

(4 Marks)

iii) Mode

(2 Marks)

QUESTION 3 **[20 MARKS]**

The results of 50 experiments recorded by a researcher are shown below.

38.13	40.24	41.24	41.68	42.31	42.82	43.37	43.86	44.23	45.10
45.55	46.00	46.20	46.94	47.00	47.52	48.14	48.58	49.02	49.18
49.97	49.99	50.26	50.51	50.68	50.82	51.07	51.32	51.48	51.65
51.86	52.14	52.20	52.60	53.14	53.98	54.35	54.87	55.22	55.90
56.14	56.30	57.03	57.92	58.20	59.50	60.10	60.50	62.50	63.86

a) Plot data above using steam and leaf.

(5 Marks)

b) Construct a histogram for these data using the Sturgis Law.

(15 Marks)

QUESTION 4 **[5 MARKS]**

A survey found that 86% of Americans have never been a victim of violent crime. If a sample of 12 American is selected at random, find the probability that 10 or more have never been victims of violent crime.

QUESTION 5 **[10 MARKS]**

A mandatory competency test for high school sophomores has aa normal distribution with a mean of 400 and a standard deviation of 100.

a) The top 3% of students receive $500. What is the minimum score you would need to receive this award?

(5 Marks)

b) The bottom 1.5% if students must go to summer school. What is the minimum score you would need to stay out of this group?

(15 Marks)